We are mathematicians: all 8 practices

I can… use mathematical habits to make sense, solve, and explain.

1 Understand & keep trying

Read, retell, and try a plan.

If I get stuck, I try another way.

MP.1

2 Connect numbers to meaning

Tell what each number counts.

12 ÷ 3 means 12 objects in 3 equal groups.

MP.2

3 Explain & listen

Show why an idea works.

"I agree because…" / "Can you show…?"

MP.3

4 Show the situation

Use a drawing, equation, or graph.

Match every part to the story.

MP.4

5 Choose a helpful tool

Pick a tool and tell why it helps.

A number line can compare fractions.

MP.5

6 Be clear & precise

Use correct words, symbols, and units.

Label 24 cm² as area, not length.

MP.6

7 Look for structure

Find useful parts and relationships.

7 × 6 = (5 × 6) + (2 × 6).

MP.7

8 Notice repeated reasoning

Look for what keeps happening.

Each ×10 makes every digit's value 10 times as much.

MP.8

How we solve a math problem

I can… represent a problem, solve it, and check my answer.

Understand the story

There are 4 packs of markers. Each pack has 6 markers. How many markers are there?

Represent → solve



4 × 6 = 24 markers

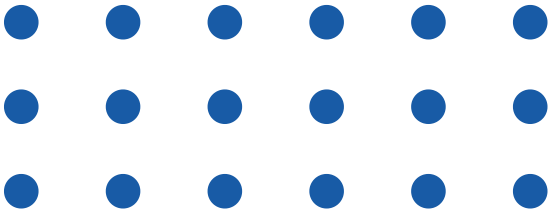
Explain → check

Four equal groups of 6 make 24. I can check: 24 ÷ 4 = 6. My answer counts markers, not packs.

Connect the model and equation

I can… explain how my drawing matches my equation.

Objects → drawing → equation



3 rows

6 in each

3 rows × 6 in each row = 18 dots

Keep the meaning

Each row is one equal group. Every dot is one object. Counting all the dots gives the product.

Use a related equation

18 ÷ 3 = 6

18 objects shared into 3 equal rows gives 6 in each row.

How we explain and discuss math

I can… explain my thinking and respond to another mathematician.

Show a reason

$6 \times 7 = (5 \times 7) + (1 \times 7) = 35 + 7 = 42$

"I broke 6 groups into 5 groups and 1 more group."

Connect to someone's idea

"I agree because your two parts still make 6 groups."
"Where do I see the extra group in your model?"

Disagree with a reason

$6 \times 7 \neq 35 + 1$

"One more group of 7 adds 7, not 1."

Our math work routines

I can… start, work with a partner, and get help independently.

Start ready

1. Sit down and take out your notebook and pencil.
2. Read the posted problem.
3. Start silently. Show your thinking.

Work with a partner

Think first → take turns explaining → compare models.
Both partners write. Both partners can explain.

When I am stuck

Reread → draw what I know → use a tool → ask a specific question.
"I know the total. I need help finding the group size."

Finish

Check your work. Put tools back. Complete the exit ticket independently.

Multiplication counts equal groups

I can... explain the factors and product in an equal-group model.

Four groups of 3



$4 \times 3 = 3 + 3 + 3 + 3 = 12$

Name what each number counts

4 counts the groups. 3 counts objects in each group. 12 counts all the objects.

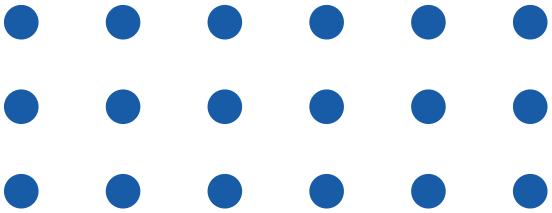
The groups must be equal

Groups of 3, 3, 2, and 4 have 12 objects, but do not model four equal groups of 3.

Arrays show multiplication

I can... use rows and columns to represent multiplication.

Three rows of 6



3 rows

6 in each

$3 \times 6 = 18$

Turn the array

Rotate it: now there are 6 rows with 3 in each. The number of objects stays 18.

$6 \times 3 = 18$

Explain the property

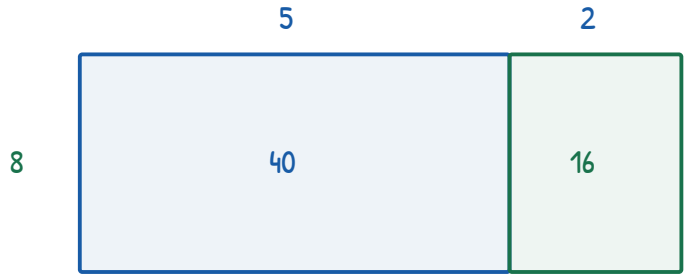
$3 \times 6 = 6 \times 3$

Changing the order of factors does not change the product.

Break apart a factor

I can... use facts I know to find a new product.

Split 7 groups into 5 + 2



7 × 8 = (5 × 8) + (2 × 8)

Combine the two products

40 + 16 = 56

Keep the groups equal

Both parts still use groups of 8. Splitting a factor does not change the total number of objects.

Division: share or make groups?

I can... explain two meanings of a division equation.

Share 12 among 3 groups



12 ÷ 3 = 4 objects in each group

Make groups of 3



12 ÷ 3 = 4 groups

Same equation, different meaning

The story tells us whether 3 is the number of groups or the size of each group. Label the answer.

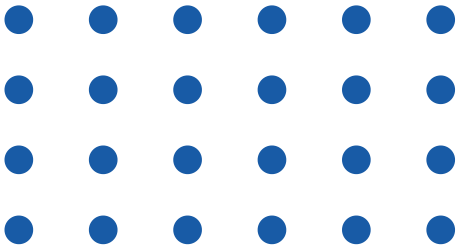
Solve a one-step group problem

I can… draw a model and equation to solve a multiplication or division story.

Read and label

There are 24 chairs. Put 6 chairs in each row. How many rows?

Make equal rows



4 rows

6 in each

r × 6 = 24 → 24 ÷ 6 = 4

Answer in the story

There are 4 rows. Check: 4 rows of 6 chairs make 24 chairs.

Division finds an unknown factor

I can… use multiplication to solve an unknown-factor problem.

What factor is missing?

$6 \times ? = 42$

Think multiplication to divide

$42 \div 6 = ? \rightarrow 6 \times 7 = 42$

The missing factor is 7.

Check the relationship

$42 \div 7 = 6$

Multiplication and division describe the same equal groups from different starting points.

Fact strategies: doubles, fives, nines

I can... use known multiplication facts to recall related facts.

Double a known fact

$2 \times 7 = 14 \rightarrow 4 \times 7 = 28$

Use half of ten groups

$10 \times 6 = 60 \rightarrow 5 \times 6 = 30$

Take one group from ten

$9 \times 8 = (10 \times 8) - 8 = 72$

Use the fact to divide

$72 \div 9 = 8$

Zero, one, and unknown numbers

I can... explain zero and one facts and solve equations with a missing number.

Zero groups and one group

$0 \times 8 = 0 \cdot 1 \times 8 = 8$

Zero equal groups contain no objects. One group contains exactly its group size.

An unknown can be anywhere

$? \times 6 = 30 \rightarrow ? = 5$

$30 = 6 \times ? \rightarrow ? = 5$

$30 \div ? = 6 \rightarrow ? = 5$

Equality means the same value

Both sides of the equal sign must represent the same amount.

Two steps: find the missing amount

I can... solve a two-step problem using addition, subtraction, or multiplication.

Connect the quantities

I need 30 stickers. I buy 4 packs with 6 in each pack. How many more stickers do I need?

First find the stickers bought



4 × 6 = 24 stickers

Then find the difference

24 + s = 30 → s = 6 stickers

Patterns in a multiplication table

I can… describe and explain a multiplication-table pattern.

Compare two rows

Factor	1	2	3	4	5
× 2	2	4	6	8	10
× 4	4	8	12	16	20

Notice and explain

The × 4 row has twice each value in the × 2 row. Four groups is twice two groups.

$4 \times 5 = 2 \times (2 \times 5) = 20$

Another pattern

Products with a factor of 5 end in 0 or 5. Every other one is also a multiple of 10.

Add by decomposing place values

I can... decompose numbers to add within 1,000.

Break apart 356 + 278

(300 + 50 + 6) + (200 + 70 + 8)

Combine like units

500 + 120 + 14 = 634

120 is 1 hundred and 2 tens. 14 is 1 ten and 4 ones.

Estimate and check

360 + 280 ≈ 640

634 – 278 = 356

Subtract by renaming place values

I can... rename and decompose numbers to subtract within 1,000.

Rename 503

503 = 400 + 90 + 13

Subtract 268 by units

(400 – 200) + (90 – 60) + (13 – 8)

200 + 30 + 5 = 235

Check by addition

235 + 268 = 503

Estimate with nearby friendly numbers

I can... estimate a sum or difference and assess whether an answer is reasonable.

Locate the number



Choose an estimate

$248 + 176 \approx 250 + 180 = 430$

Compare to the exact answer

$248 + 176 = 424$

424 is close to 430. Estimates depend on the purpose; they do not have to equal the exact answer.

Multiply a one-digit number by tens

I can... use place value to multiply a one-digit number by a multiple of ten.

Four groups of 30



Count tens first

4 × 3 tens = 12 tens

Name the value

12 tens = 120 → 4 × 30 = 120

The zero is a result of counting tens, not a trick added to a fact.

A unit fraction is one equal part

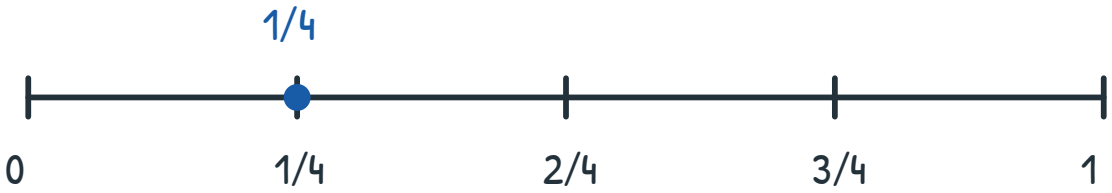
I can… represent a unit fraction with an area model and a number line.

One out of four equal parts



The full strip is one whole. One equal part is one fourth.

One fourth of a unit length



Name the unit

A unit fraction has numerator 1. Its denominator names how many equal parts make the whole.

Fractions count unit-fraction lengths

I can... explain the numerator by counting unit fractions from zero.

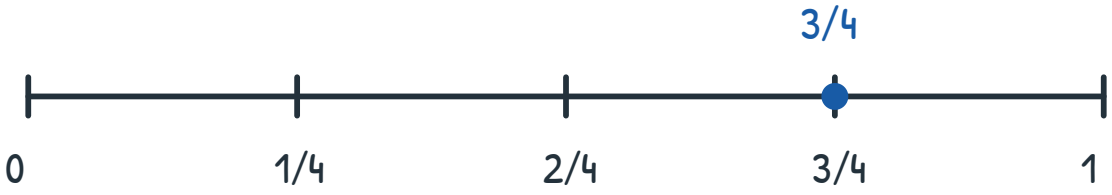
Three lengths of one fourth

3/4



3/4 = three units of 1/4

Locate the same number



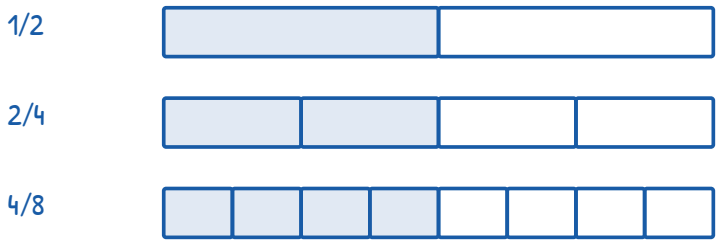
Count intervals

The numerator counts unit lengths from 0. Count three spaces of length 1/4, not three tick marks.

Equivalent fractions have equal value

I can... explain equivalent fractions using related partitions.

Repartition the same whole



Explain why

1/2 = 2/4 = 4/8

Each half becomes two fourths or four eighths. More smaller parts cover the same amount.

Also connect thirds and sixths

1/3 = 2/6

Whole numbers can be fractions

I can... use a model to express a whole number as a fraction.

Each whole contains fourths

1 whole 

1 whole 

2 = 8/4

A whole is all its equal parts

4/4 = 1 · 6/6 = 1

Another way to name two

2 = 2/1 = 4/2 = 8/4

The value stays two, even when the unit used to count it changes.

Compare fractions with matching parts

I can... compare fractions with a common numerator or denominator.

Same denominator

3/8



5/8



3/8 < 5/8

Same numerator

1/2 > 1/4

One half is larger than one fourth of the same whole.

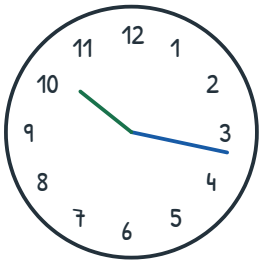
Explain the reason

Same denominator: compare the number of equal-size parts.
Same numerator: compare the size of the parts.

Read time to the nearest minute

I can... read and write time to the nearest minute.

Read the hour, then minutes



Hour hand: after 10

Minute hand: 17

Count minutes from 12

3 groups of 5 + 2 = 17 minutes

Write the time

10:17

The hour hand is between 10 and 11 because the hour is not finished.

Elapsed time within one hour

I can… find an elapsed time within the same hour.

Read the start and end

Reading starts at 9:12 and ends at 9:37.

Count forward

9:12 → 9:32 → 9:37

+20 minutes + 5 minutes = 25 minutes

Find another unknown

Start: 9:12. Duration: 25 minutes.

End time: 9:37

Measure customary lengths

I can... measure length to the nearest half or quarter inch.

Read equal partitions



Start at zero

Place one end of the object at 0 on the ruler. Read the mark at the other end. A pencil ending at 4 1/4 measures 4 1/4 inches.

Choose an appropriate unit

Use inches for a pencil, feet for a room, and yards for a large field.

Customary weight and capacity

I can… measure and solve a one-step customary measurement problem.

Choose the attribute and unit

Attribute	Units
Weight	ounces, pounds
Capacity	cups, pints, quarts, gallons

Solve with the same unit

Each jug holds 4 cups. Three jugs hold how much?



3 × 4 = 12 cups

Label what you measured

Cups describe capacity. Pounds describe weight. They answer different questions.

Survey questions and frequency tables

I can… collect categorical data and organize counts in a table.

Ask one clear question

Which fruit is your favorite: apple, banana, orange, or grape? Each student votes once.

Organize the votes

Fruit	Votes
Apple	8
Banana	6
Orange	4
Grape	2

Check and interpret

8 + 6 + 4 + 2 = 20 students

Apple has the most votes. Apple has 2 more votes than banana.

Read a scaled picture graph

I can… use a picture-graph key to find and compare counts.

Favorite fruit

Each circle represents 2 votes.



Use the key, not just the pictures

Apple: 4 × 2 = 8 votes

Apple and orange: 8 + 4 = 12 votes

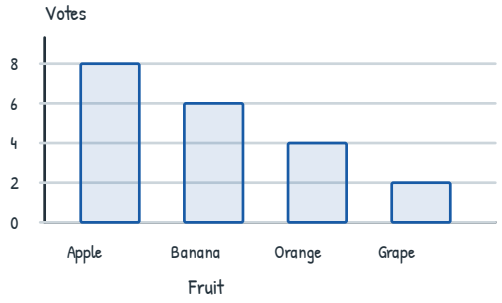
Compare

Apple – banana: 8 – 6 = 2 more votes

Read a scaled bar graph

I can... use a bar graph's scale to answer questions.

Favorite fruit



Check the scale

The vertical labels count by 2. A bar reaching 6 represents 6 votes, not 3 votes.

A two-step comparison

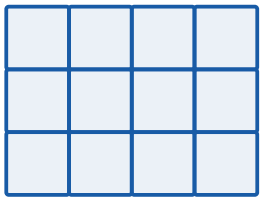
(Banana + orange) − apple

(6 + 4) − 8 = 2 votes

Area counts unit squares

I can… find a rectangle's area by counting unit squares.

Tile without gaps or overlaps



4 columns × 3 rows

Count square units

4 + 4 + 4 = 12 square units

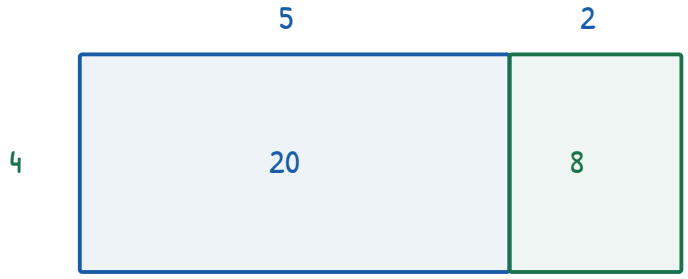
Name the attribute

Area measures how much surface is covered. Each tile covers one square unit.

Multiply and split rectangular areas

I can... connect rectangular area to multiplication and addition.

Split a 7 by 4 rectangle



Combine the parts

7 × 4 = (5 × 4) + (2 × 4)

20 + 8 = 28 square units

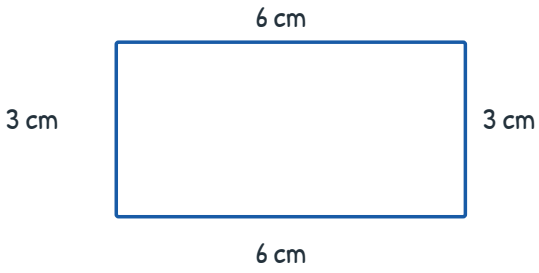
Check the full rectangle

Seven columns of four squares contain the same 28 squares. The split changes the calculation, not the area.

Perimeter follows the boundary

I can... add side lengths to find a polygon's perimeter.

Trace every side once



Add lengths, not square units

6 + 3 + 6 + 3 = 18 cm

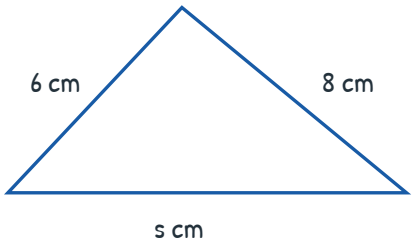
Explain the meaning

18 centimeters measures the distance around the rectangle. It does not measure the inside.

Find an unknown side length

I can... use perimeter and known sides to find a missing side.

A triangle has perimeter 20 cm



Write an equation

6 + 8 + s = 20

Find what is left

s = 20 – 14 = 6 cm

Check: 6 + 8 + 6 = 20 cm.

Classify quadrilaterals by attributes

I can... use attributes to classify quadrilaterals.

Start with four sides

A quadrilateral is a polygon with four sides.

Use more specific attributes

Shape	Attributes
Rectangle	4 right angles
Rhombus	4 equal sides
Square	4 right angles AND 4 equal sides

Categories can overlap

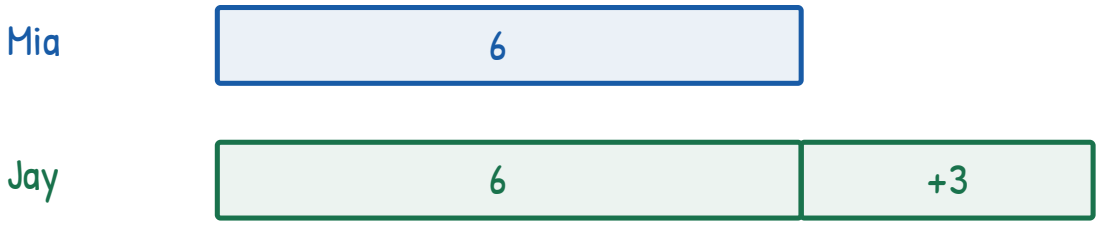
A square is a rectangle and a rhombus. Each of these shapes is also a quadrilateral.

Additive or multiplicative comparison?

I can... distinguish "more than" from "times as many."

Additive: a difference

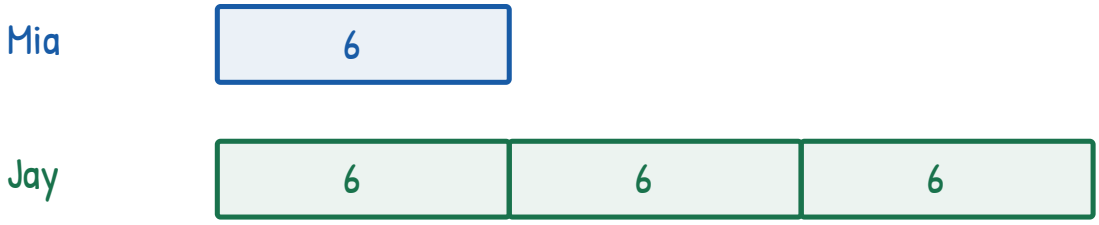
Mia has 6 beads. Jay has 3 more beads than Mia.



6 + 3 = 9 beads

Multiplicative: equal copies

Mia has 6 beads. Jay has 3 times as many beads.



3 × 6 = 18 beads

What is unknown in a comparison?

I can... solve a multiplicative comparison with any quantity unknown.

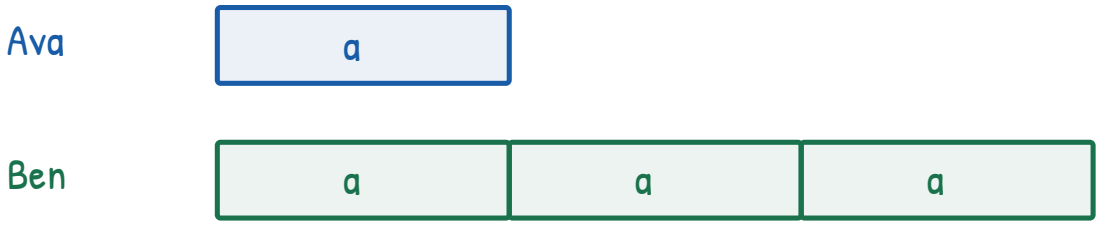
Larger amount unknown

Ava has 5 cards. Ben has 3 times as many.

$3 \times 5 = b \rightarrow b = 15$ cards

Smaller amount unknown

Ben has 15 cards, 3 times Ava's amount.



$3 \times a = 15 \rightarrow a = 5$ cards

Multiplier unknown

Ben has 15 cards. Ava has 5. How many times as many does Ben have?

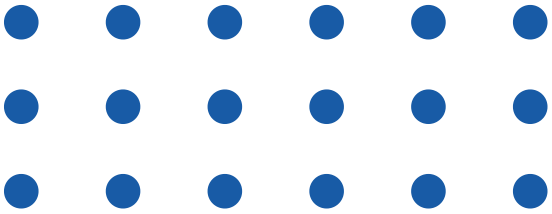
$m \times 5 = 15 \rightarrow m = 3$

Multiplication and division situations

I can… tell whether the total, group size, or number of groups is unknown.

Total unknown

3 bags hold 6 apples each.



3 rows

6 in each

3 × 6 = 18 apples

Group size unknown

18 apples are shared equally into 3 bags.

3 × a = 18 → 18 ÷ 3 = 6 apples per bag

Number of groups unknown

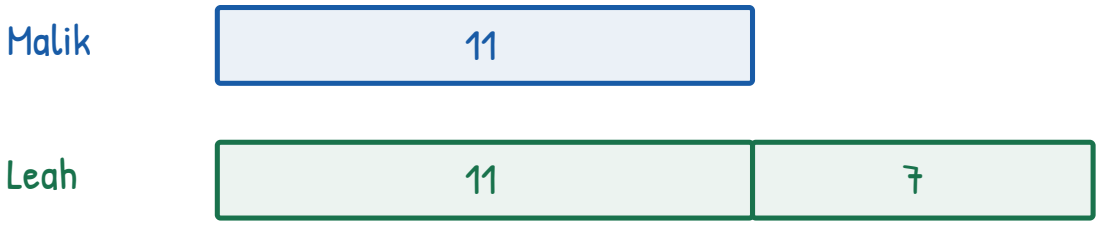
Pack 18 apples with 6 in each bag.

b × 6 = 18 → 18 ÷ 6 = 3 bags

Additive comparison: the smaller amount is unknown

I can... draw an additive comparison to find a smaller amount.

Leah has 18 stickers. She has 7 more than Malik. How many does Malik have?



m + 7 = 18

18 – 7 = 11 stickers

Malik has 11. Leah has 7 more: 11 + 7 = 18.

Additive comparison: find the difference

I can… represent the difference between two quantities.

Nia reads 24 pages. Luis reads 17 pages. How many more pages does Nia read?



17 + d = 24

d = 24 – 17 = 7 pages

The equal parts match. The extra part is the difference, not the total.

Check your answer

I can… use estimation and an inverse operation to check an answer.

Make an estimate

A library has 248 fiction books and 176 nonfiction books. How many books?

250 + 180 = 430 → about 430 books

Calculate and check

248 + 176 = 424

424 − 176 = 248

Does it fit the story?

424 is close to 430. The total is greater than either part. My answer is 424 books.

Solve a two-step problem

I can… represent and solve a problem that needs two operations.

Read for relationships

We have 5 boxes of pencils with 12 in each box. We give away 18 pencils. How many remain?

Find what we need first

5 boxes



5 × 12 = 60 pencils

Then answer the question

60 – 18 = p → p = 42 pencils

Estimate: 60 – 20 = 40. An answer of 42 is reasonable.

What does the remainder mean?

I can… interpret a remainder using the situation.

One calculation

29 ÷ 4 = 7 R1

4 × 7 + 1 = 29

Different questions → different answers

Situation	Answer
29 students; 4 seats per car. Cars needed?	8 cars
29 beads; 4 per full bracelet. Full bracelets?	7 bracelets
29 cards shared by 4 players. Cards each and left over?	7 each; 1 left

Explain your decision

The extra student needs a seat. The extra bead does not make a full bracelet. The question tells us what to do with the leftover.

Write a letter equation for two steps

I can… represent and solve a two-step story with an unknown.

There are 96 crayons. Four tables share them equally. Each table uses 5 crayons. How many crayons remain at each table?

$c = (96 \div 4) - 5$

$96 \div 4 = 24$

$c = 24 - 5 = 19$ crayons

$19 + 5 = 24$; $4 \times 24 = 96$

The answer is per table, so do not subtract 5 from 96 first.

Factors, multiples, prime, composite

I can… find factor pairs and use them to classify a number.

Factor pairs make a product

Factor pair for 24	Product
1 × 24	24
2 × 12	24
3 × 8	24
4 × 6	24

Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24.

Multiples count equal groups

Multiples of 6: 6, 12, 18, 24, 30, …

24 is a multiple of 6 because 6 × 4 = 24.

Classify by factors

13: only 1 and 13 → prime.
24: more than two factors → composite.
1 has one factor → neither.

Find every factor pair

I can… list factor pairs systematically.

First factor	Matching factor
1	24
2	12
3	8
4	6

Each row makes a rectangle with 24 unit squares.

1×24, 2×12, 3×8, 4×6

Test 5: it does not divide 24 evenly. At 6, the pair 6×4 repeats a pair we already found.

Factors: 1, 2, 3, 4, 6, 8, 12, 24

Factors and multiples are connected

I can… tell whether a number is a multiple of a one-digit number.

Can 42 counters make equal groups of 6?

6, 12, 18, 24, 30, 36, 42

$6 \times 7 = 42$

6 and 7 are factors of 42. The product 42 is a multiple of both.

42 is a multiple of 6.

43 is not: 43 ÷ 6 leaves 1.

Prime, composite, or neither?

I can… classify a number by its positive factors.

Number	Factor pairs
13	1 × 13
12	1 × 12; 2 × 6; 3 × 4
1	1 × 1

13: exactly 2 factors → prime.
12: more than 2 factors → composite.

1 has only one factor, so it is neither. 2 is prime even though it is even.

Generate and explain patterns

I can… generate a pattern and explain a feature beyond its rule.

Number pattern

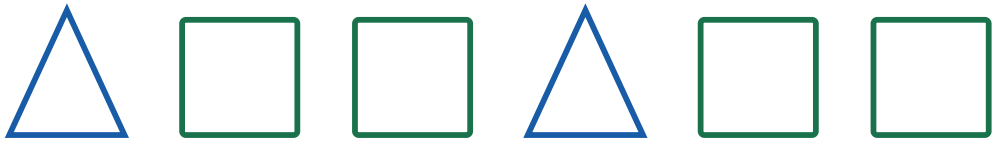
Start at 2. Add 3 each time.

2 → 5 → 8 → 11 → 14 → 17

A feature: even and odd terms alternate. Adding an odd number changes parity.

Shape pattern

Repeat: triangle, square, square.



Positions 1, 4, 7, 10… are triangles. Each repeat has 3 shapes.

Each place is ten times as much

I can… explain how a digit's value changes in adjacent places.

Same digit, different values

Number	Value of 4
4,400: first 4	4,000
4,400: second 4	400

4,000 = 10 × 400

Explain the relationship

4 thousands = 40 hundreds.
One thousand is 10 hundreds. The 4 in the thousands place has 10 times the value of the 4 in the hundreds place.

Try another pair

700 = 10 × 70

Compare values, not just the names of the places.

One number, different forms

I can… read and write a number in standard, word, and expanded form.

Place each digit

Ten-thousands	Thousands	Hundreds	Tens	Ones
4	0	6	0	8

Three ways to name it

Standard: 40,608
Word: forty thousand six hundred eight
Expanded: 40,000 + 600 + 8

Zero holds a place

There are no thousands and no tens. The zeros keep the other digits in the correct places.

Addition: regrouping keeps the value

I can... explain regrouping in the addition algorithm.

Add like units

Hundreds	Tens	Ones
3	6	8
+ 2	7	7
5	13	15

Exchange without changing the amount

15 ones = 1 ten + 5 ones.
13 tens + 1 ten = 14 tens.
14 tens = 1 hundred + 4 tens.

Record the total

368 + 277 = 645

5 hundreds + 1 regrouped hundred = 6 hundreds. The total value stays the same at every exchange.

Subtraction: rename before subtracting

I can… rename units to explain subtraction across zeros.

Rename 4,002

Thousands	Hundreds	Tens	Ones
4	0	0	2
3	10	0	2
3	9	10	2
3	9	9	12

Now subtract 1,786

Thousands	Hundreds	Tens	Ones
3	9	9	12
– 1	7	8	6
2	2	1	6

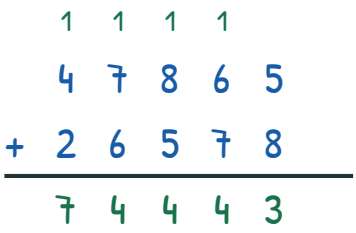
4,002 – 1,786 = 2,216

Check the value

2,216 + 1,786 = 4,002

Addition algorithm: line up the places

I can... use place value to explain the standard addition algorithm.



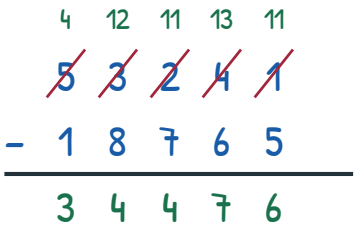
3 ones remain after exchanging 13 ones for 1 ten and 3 ones.
14 tens → 1 hundred and 4 tens.
14 hundreds → 1 thousand and 4 hundreds.
14 thousands → 1 ten-thousand and 4 thousands.

47,865 + 26,578 = 74,443

Each small green 1 is one unit of its new place, not always one single object.

Subtraction algorithm: rename, then subtract

I can… connect renamed units to a standard subtraction calculation.



Before subtracting, rename 53,241 as:
4 ten-thousands + 12 thousands + 11 hundreds + 13 tens + 11 ones.

53,241 – 18,765 = 34,476

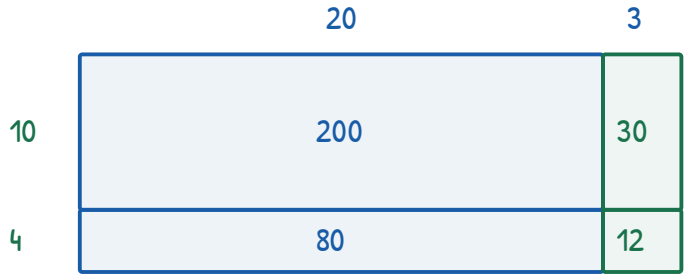
34,476 + 18,765 = 53,241

Break apart factors to multiply

I can... use an area model and partial products to multiply.

Break 23 into 20 + 3

23 × 14 = (20 + 3) × (10 + 4)



Keep every partial product

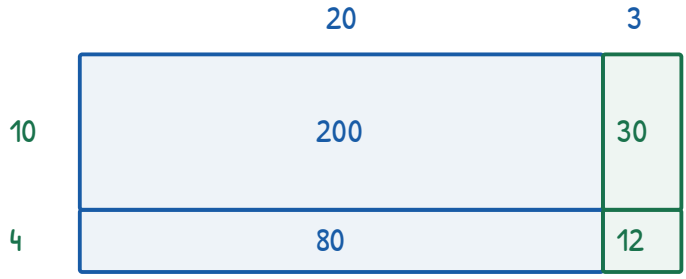
200 + 80 + 30 + 12 = 322

Each rectangle is one part of the full area. All four parts must be counted.

Connect multiplication to the algorithm

I can… explain how the multiplication algorithm records partial products.

First understand the parts



$23 \times 4 = 92 \cdot 23 \times 10 = 230$

Record them vertically

Step	Partial product
Multiply by 4 ones	92
Multiply by 1 ten	230
Add both products	322

$23 \times 14 = 322$

Why the zero?

230 counts 23 tens. The second row is multiplication by 10, not by 1.

Three-digit multiplication: each place has a job

I can… multiply a three-digit number by a one-digit number using place value.

Part of 243	Multiply by 3
200	600
40	120
3	9

243 × 3 = 600 + 120 + 9

= 729

240 × 3 = 720

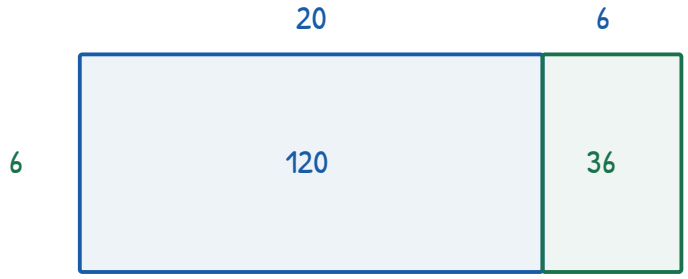
729 is close to the estimate. Every part was multiplied by 3.

Break apart the dividend to divide

I can... use partial quotients and multiplication to explain division.

156 objects in 6 equal groups

156 = 120 + 36



Take manageable groups

Amount remaining	Groups of 6
156 – 120 = 36	20
36 – 36 = 0	6

156 ÷ 6 = 20 + 6 = 26

Check with multiplication

6 × 26 = 156

Partial quotients can leave a remainder

I can... divide with partial quotients and explain the leftover.

Subtract groups of 4	Groups taken
857 – 800 = 57	200
57 – 40 = 17	10
17 – 16 = 1	4

200 + 10 + 4 = 214

857 ÷ 4 = 214 R1

4 × 214 + 1 = 857

1 is less than 4, so another complete group cannot be made.

Compare whole numbers

I can... compare whole numbers using the value of their digits.

Start with the greatest place

Place	47,305	47,350
Ten-thousands	4	4
Thousands	7	7
Hundreds	3	3
Tens	0	5

Find the first difference

47,305 < 47,350

The ten-thousands, thousands, and hundreds match. At the tens place, 0 tens is less than 5 tens.

Read the whole statement

"Forty-seven thousand three hundred five is less than forty-seven thousand three hundred fifty."

Fractions: equal parts of one whole

I can… identify the whole and explain a fraction using equal parts.

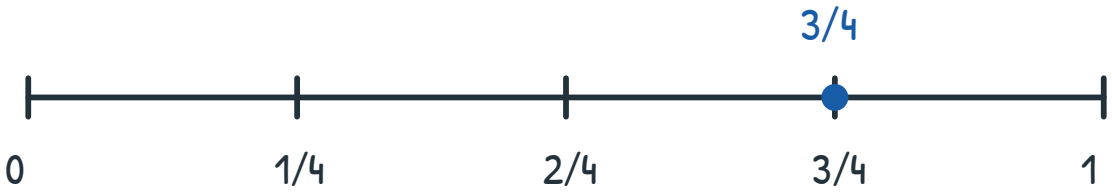
Name the whole and the unit

3/4



The entire strip is 1 whole. Each equal part is 1/4. Three of those parts make 3/4.

Fractions are numbers



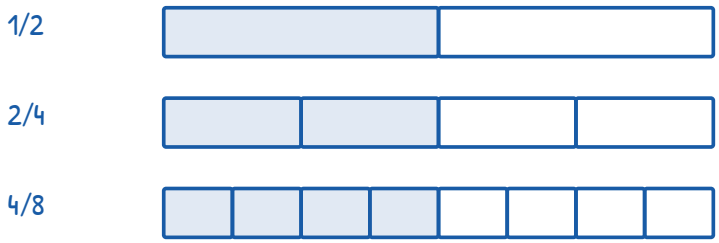
Same whole matters

A half of a small sandwich and a half of a large sandwich are different amounts of food. Name the whole before comparing.

Equivalent fractions

I can... use models to explain why fractions are equivalent.

Same whole, same shaded amount

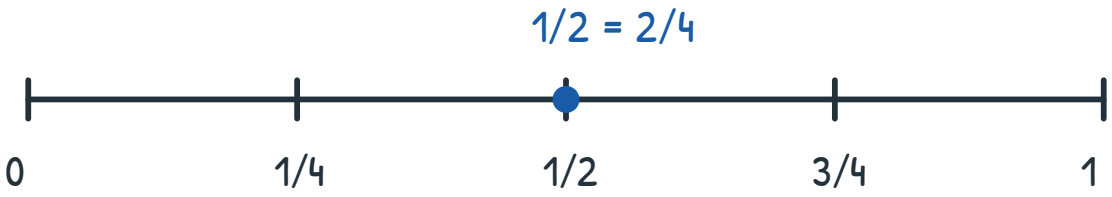


Explain what changed

Each half is split into 2 equal parts to make fourths. The number of parts doubles; each part is half as large. The amount stays the same.

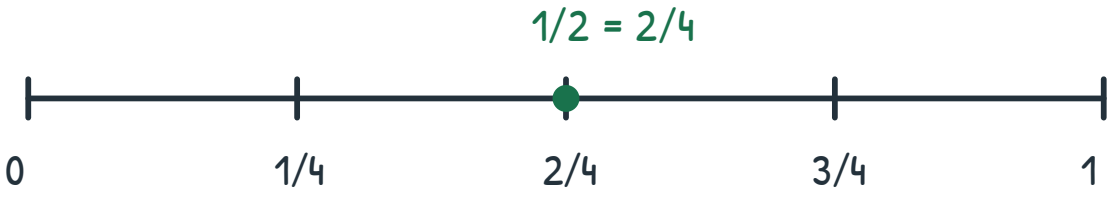
1/2 = 2/4 = 4/8

On a number line



Equivalent fractions meet at the same point

I can... show equivalent fractions on a number line.



1/2 = 2/4 = 4/8

Partition each half into 2 fourths, then each fourth into 2 eighths. The distance from 0 stays the same.

More pieces, smaller pieces, same amount. Multiply numerator and denominator by the same whole number.

Compare fractions by reasoning

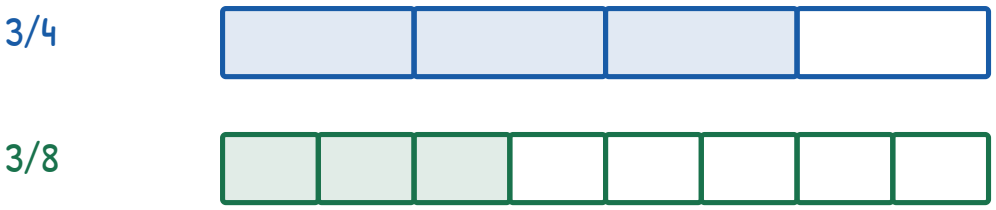
I can… compare fractions and justify my comparison with models or benchmarks.

Same denominator: count parts

5/8 > 3/8

Same-size parts; five parts is more than three.

Same numerator: compare part sizes



3/4 > 3/8

Different parts: use a benchmark

2/5 < 1/2 < 3/4

2/5 is less than half; 3/4 is greater than half. So 2/5 < 3/4. All comparisons use the same whole.

Compare fractions with a half benchmark

I can... compare fractions by locating them around one half.

3/8



5/6



$3/8 < 4/8 = 1/2$

$5/6 > 3/6 = 1/2$

$3/8 < 5/6$

The blue amount is below half; the green amount is above half. Both bars represent the same whole.

Compare fractions with the same numerator

I can... compare equal counts of different-size fractional parts.

3/4



3/8



Both amounts count 3 parts. Fourths are larger than eighths of the same whole.

3/4 > 3/8

A larger denominator means smaller unit parts. That reasoning works here because the numerators match.

Decompose and rename fractions

I can... write a fraction as a sum in more than one way.

Three ways to make 5/6

5/6



5/6 = 1/6 + 1/6 + 1/6 + 1/6 + 1/6

5/6 = 2/6 + 3/6 = 4/6 + 1/6

Rename an amount greater than one

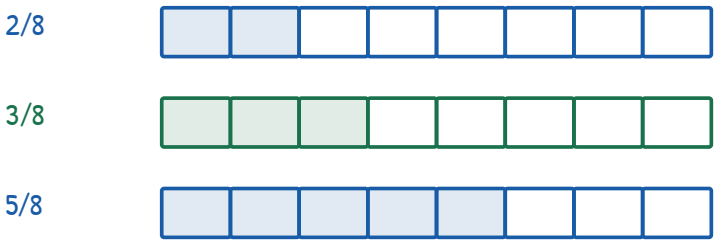
8/6 = 6/6 + 2/6 = 1 2/6

Six sixths make one whole. Two more sixths remain.

Add and subtract fractions

I can... add and subtract fractions with like denominators.

Join same-size parts



2/8 + 3/8 = 5/8

Separate same-size parts

7/8 – 3/8 = 4/8 = 1/2

Why keep the denominator?

The unit is still one eighth. We change how many eighths we have, not the size of each part.

2/8 + 3/8 ≠ 5/16

Add and subtract mixed numbers

I can... rename a whole to add or subtract mixed numbers.

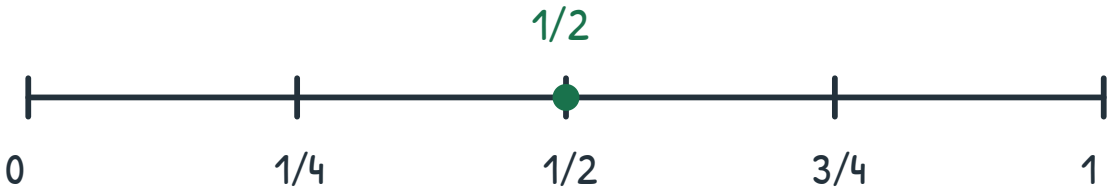
Subtract: 2 1/4 – 1 3/4

2 1/4 = 1 + 4/4 + 1/4 = 1 5/4

Rename one whole as four fourths. The amount is still 2 1/4.

Now subtract

1 5/4 – 1 3/4 = 2/4 = 1/2



Addition may make a whole

1 3/4 + 2 2/4 = 3 5/4 = 4 1/4

Fractions can be greater than one

I can... connect a fraction greater than one to a mixed number.



7/4 = 4/4 + 3/4

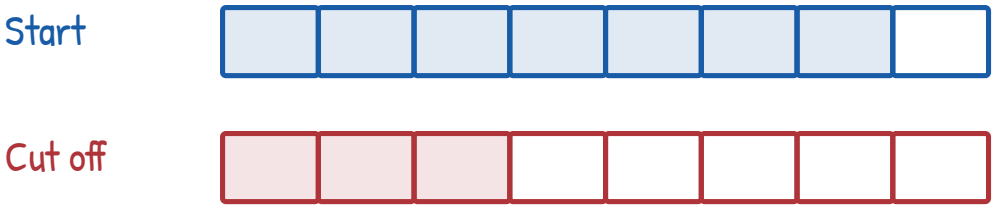
7/4 = 1 3/4

The denominator names the size of each part. Seven fourths pass one whole but do not reach two.

Fraction word problems: join or separate?

I can... use a visual model to write a fraction equation.

A ribbon is 7/8 meter long. Cut off 3/8 meter. How much remains?



7/8 − 3/8 = 4/8

4/8 = 1/2 meter remains

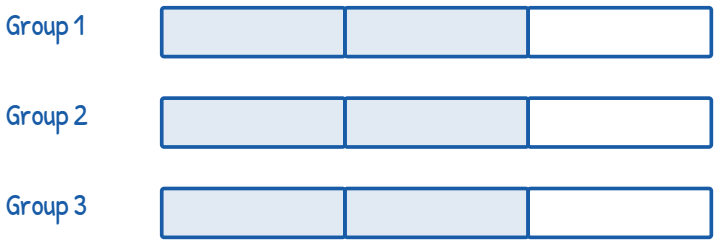
3/8 + 4/8 = 7/8

The denominator stays 8 because each part is still one eighth of a meter.

Multiply a whole number and a fraction

I can... use equal groups to explain whole-number multiplication with a fraction.

Three groups of 2/3



$3 \times 2/3 = 2/3 + 2/3 + 2/3 = 6/3 = 2$

A fraction of a set

1/4 of 12 counters: split 12 into 4 equal groups. One group has 3.

$12 \times 1/4 = 3$

Connect to a story

Three ribbons are each 2/3 meter long. Their total length is 2 meters.

Multiply a fraction in a story

I can... model repeated groups of a fraction.

Each badge uses 2/3 meter of ribbon. Make 4 badges.

2/3 + 2/3 + 2/3 + 2/3

4 × 2/3 = 8 × 1/3 = 8/3

2 2/3 meters of ribbon

Four groups of two thirds contain eight thirds. The denominator continues to name thirds.

Tenths, hundredths, and decimals

I can... connect decimal notation to tenths and hundredths.

Same value in different units

4/10



4/10 = 40/100 = 0.4 = 0.40

Add tenths and hundredths

3/10 + 25/100 = 30/100 + 25/100

= 55/100 = 0.55

Explain the zero

4 tenths is 40 hundredths. The zero in 0.40 does not change the value. Ten hundredths make one tenth.

Add tenths and hundredths as fractions

I can… rename tenths as hundredths before adding.

Walk 3/10 kilometer, then 24/100 kilometer.

3/10 = 30/100

30/100 + 24/100 = 54/100

54/100 kilometer = 0.54 kilometer

The 3 tenths were renamed, not changed. Count like-size parts before joining them.

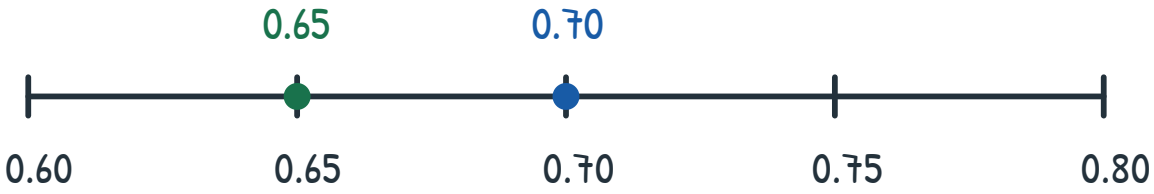
Compare decimals by value

I can... compare decimals to hundredths and justify their sizes.

Use the same whole

0.7 = 70/100 • 0.65 = 65/100

Locate both numbers



Explain the comparison

0.7 > 0.65

70 hundredths is greater than 65 hundredths. More written digits does not mean a greater number.

0.65 is not greater just because 65 > 7.

Choose and use metric units

I can... choose a metric unit and solve a measurement problem.

Choose a useful unit

Measure	Smaller unit	Larger unit
Length	centimeter (cm)	meter (m)
Mass	gram (g)	kilogram (kg)
Capacity	milliliter (mL)	liter (L)

Picture the size

A doorway is about 2 m high.
A small paper clip has a mass of about 1 g.
A large drink bottle may hold 1 L.

Solve using the same unit

Four bottles each hold 250 mL.

4 × 250 = 1,000 mL

Metric measurement stories use matching units

I can… solve a one-step metric measurement problem.

A jug holds 1,200 mL. Pour out 375 mL. How much remains?



1,200 − 375 = 825

825 mL remains

825 + 375 = 1,200 mL

Both measurements use milliliters, so the difference also uses milliliters.

Convert larger metric units to smaller

I can... use multiplication to convert a larger metric unit to a smaller unit.

Know the unit relationship

1 m = 100 cm

1 kg = 1,000 g • 1 L = 1,000 mL

Build a conversion table

Meters	Centimeters
1	100
3	300
8	800

3 m = 3 × 100 cm = 300 cm

Explain the size

The length stays the same. Smaller units require a greater number of units to measure that length.

A length model explains metric conversion

I can... use a length model to convert meters to centimeters.



Every 1-meter section measures 100 centimeters.

$3 \times 100 = 300$

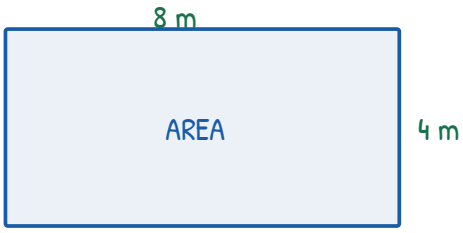
3 meters = 300 centimeters

The length stays the same. More smaller units are needed to measure it.

Area and perimeter measure different things

I can… explain and calculate a rectangle's area and perimeter.

Inside versus boundary



PERIMETER follows all 4 sides

Area counts square units

$A = \text{length} \times \text{width} = 8 \times 4 = 32 \text{ m}^2$

Perimeter counts length around

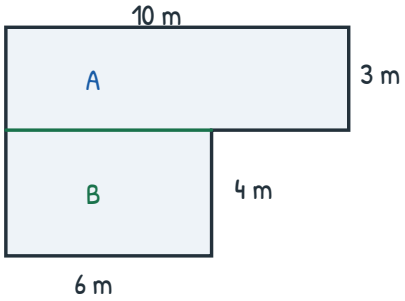
$P = 8 + 4 + 8 + 4 = 24 \text{ m}$

Area uses square meters. Perimeter uses meters.

Area of a rectilinear figure

I can… split a rectilinear figure into rectangles to find its area.

Split without overlapping



Find each rectangle's area

A: $10 \times 3 = 30$ m²

B: $6 \times 4 = 24$ m²

Combine the parts

$30 + 24 = 54$ m²

The two parts cover the shape exactly once.

Same area, different perimeter

I can... compare rectangles with a fixed area or a fixed perimeter.

Keep the area at 24 square units

Dimensions	Area	Perimeter
1 × 24	24	50
2 × 12	24	28
3 × 8	24	22
4 × 6	24	20

Keep the perimeter at 20 units

Dimensions	Perimeter	Area
1 × 9	20	9
2 × 8	20	16
4 × 6	20	24

What do you notice?

Equal areas can have different perimeters. Equal perimeters can enclose different areas.

Same perimeter can enclose different areas

I can… compare rectangles with a fixed perimeter.

Length × width	Perimeter	Area
1 × 7	16 units	7 square units
2 × 6	16 units	12 square units
3 × 5	16 units	15 square units
4 × 4	16 units	16 square units

P = 2 × (length + width)

Each pair of dimensions adds to 8, so each perimeter is 16.

All have the same boundary length. The 4-by-4 rectangle covers the greatest area of these choices.

Ask a question and organize data

I can... distinguish data types and organize numerical data in a frequency table.

What kind of answer?

Question	Data type
What is your favorite fruit?	Categorical: names
How many pets do you have?	Numerical: counts

Collect whole-number data

Pet counts: 0, 1, 2, 1, 0, 3, 2, 1, 1, 2

Count each value

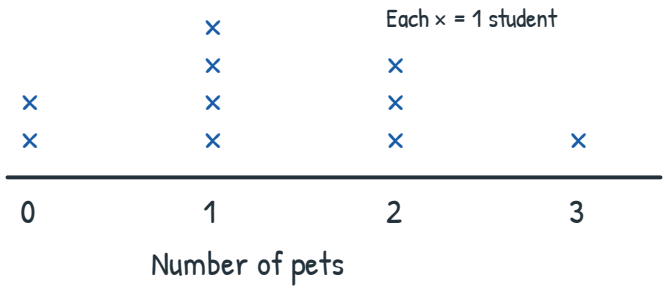
Pets	Frequency
0	2
1	4
2	3
3	1

2 + 4 + 3 + 1 = 10 students

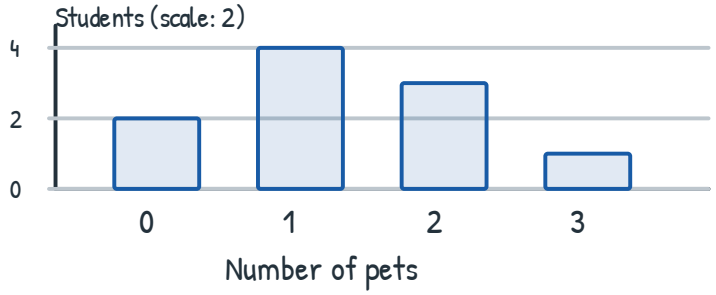
Create and read graphs

I can… represent whole-number data and answer questions about frequency.

Pets in our class: line plot



Same data: bar graph



Read and compare

4 students have 1 pet. 3 have 2 pets.
How many more have 1 pet? 4 – 3 = 1 student.

Build a frequency table from numerical data

I can… organize whole-number data in a frequency table.

Number of pets: 0, 1, 2, 1, 1, 0, 2, 1, 2, 1, 2, 1.

Number of pets	Frequency
0	2
1	6
2	4

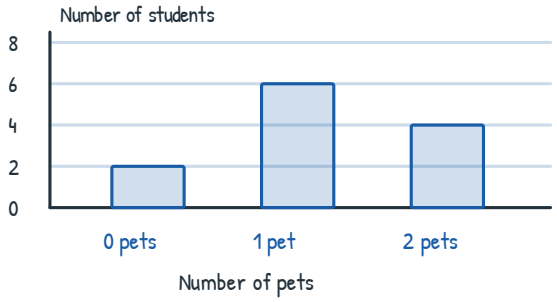
2 + 6 + 4 = 12 students

Frequency counts how often each value occurs.

"How many pets do you have?" gives numerical data. "Which pet do you like?" gives categorical data.

Create a scaled bar graph

I can… choose and label a scale to represent whole-number data.



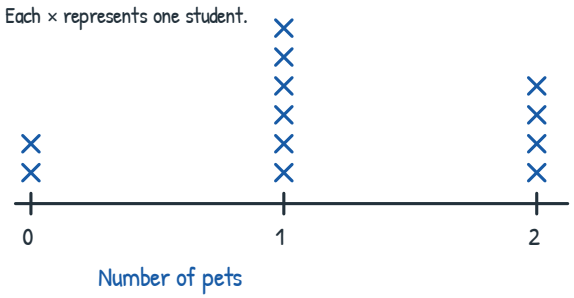
The vertical scale counts by 2. A bar at 6 means six students, not three students.

6 − 2 = 4 more students

There are 4 more students with 1 pet than with 0 pets. Include a title and labels for both axes.

Create and read a whole-number line plot

I can… use one mark per observation on a line plot.



Stack an × above the value for each student. The tallest stack is at 1 pet.

2 + 6 + 4 = 12 observations

Six students have 1 pet. Four have 2 pets. Counting marks gives students; reading the axis gives pets.

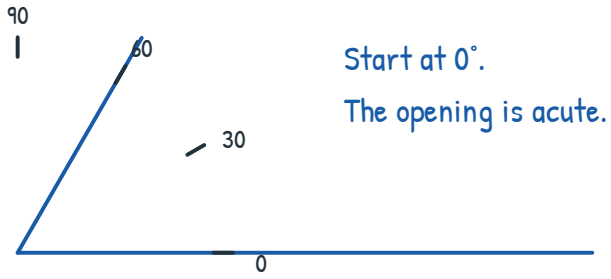
Measure, draw, and find missing angles

I can… measure and draw angles and find an unknown angle using addition or subtraction.

Use a protractor carefully

1. Put its center on the vertex.
2. Align the zero line with one ray.
3. Start at 0 on that ray's scale.
4. Read where the other ray crosses.

Measure and sketch 60°



Start at 0°.

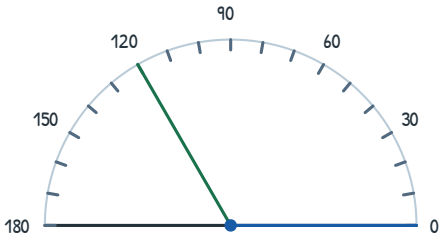
The opening is acute.

An angle is made of parts

$35^\circ + x = 90^\circ \rightarrow x = 55^\circ$

Use the protractor scale that begins at zero

I can… align a protractor and measure an angle.



Start at 0 on the first ray.

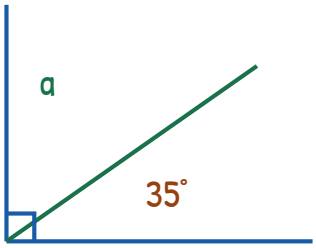
1. Put the center on the vertex.
2. Put the zero line along the first ray.
3. Read the scale that starts at that zero.

The angle measures 120°.

The opening is wider than a right angle, so 120° fits. 60° would be acute.

Find an unknown part of an angle

I can... use addition or subtraction to find an unknown angle.



90° total

A right angle is split into 35° and an unknown part.

$35^\circ + a = 90^\circ$

$a = 90^\circ - 35^\circ = 55^\circ$

$35^\circ + 55^\circ = 90^\circ$

The parts share a vertex and do not overlap.

Elapsed time across the hour

I can… find elapsed time, start time, or end time across an hour.

Find the elapsed time

A club starts at 2:45 p.m. and ends at 4:10 p.m.



15 + 60 + 10 = 85 min = 1 h 25 min

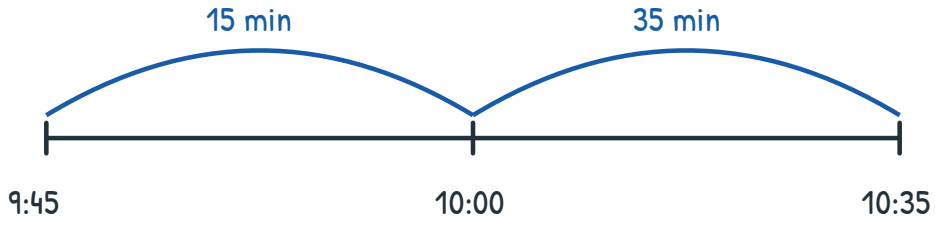
Find another unknown

End time 4:10; duration 1 h 25 min.
Count back 1 hour to 3:10, then 25 minutes to 2:45.

Elapsed time: find the ending time

I can... find an end time when an interval crosses an hour.

Start reading at 9:45 a.m. Read for 50 minutes.



50 = 15 + 35 minutes

End time: 10:35 a.m.

9:45 to 10:00 is 15 minutes. Add the remaining 35 minutes after the hour.

Elapsed time: find the starting time

I can… count back to find a start time.

A lesson ends at 1:20 p.m. It lasts 45 minutes.

Count back	Time
20 minutes	1:00 p.m.
25 more minutes	12:35 p.m.

20 + 25 = 45 minutes

Start time: 12:35 p.m.

Check forward: 12:35 to 1:00 is 25 minutes; 1:00 to 1:20 is 20 minutes.

Points, lines, segments, and rays

I can... draw and identify geometric objects and line relationships.

Endpoints tell the difference

Point • A

Segment

Ray 

Line 

Line relationships

Parallel: coplanar lines that never meet.
Perpendicular: lines that meet at right angles.





Angles measure the opening

I can… classify an angle by the amount of turn between its rays.

Compare to a right angle



Acute: < 90°

Right: 90°

Obtuse: 90°–180°

Straight and full turns

Straight angle = 180° · Full turn = 360°

Ray length does not change the angle

An angle's measure depends on the opening between the rays. Extending either ray does not make the angle larger.

Parallel and perpendicular lines

I can... identify lines by whether and how they meet.



Parallel

Perpendicular

Parallel: same distance apart; never meet.
Perpendicular: intersect to make right angles.

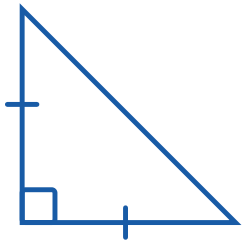
Look at the opposite sides and adjacent sides of a rectangle. Turning the drawing does not change the relationship.

Classify triangles and quadrilaterals

I can… classify shapes using side lengths, angles, and line relationships.

Triangles have two descriptions

Sides: equilateral, isosceles, or scalene.
Angles: acute, right, or obtuse.
A triangle may be both isosceles and right.



2 equal sides

1 right angle

Quadrilateral categories overlap

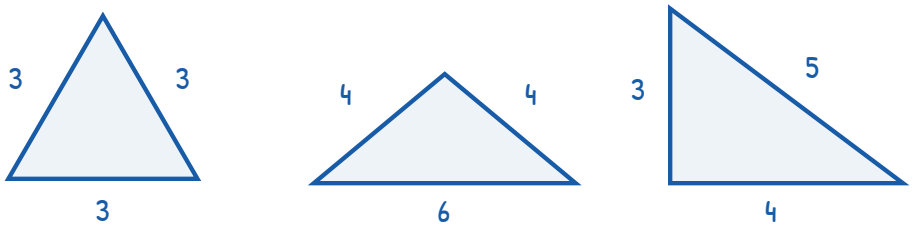
A parallelogram has two pairs of parallel sides.
A rectangle has four right angles.
A rhombus has four equal sides.
A square has both!

Explain a square's memberships

Square → rectangle AND rhombus → parallelogram

Name triangles by their side lengths

I can... classify a triangle by its equal sides.



Equilateral Isosceles Scalene

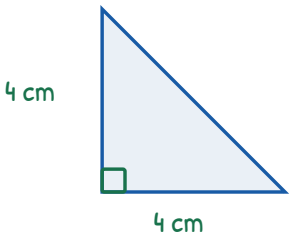
Triangle type	Side lengths
Equilateral	3 cm, 3 cm, 3 cm
Isosceles	4 cm, 4 cm, 6 cm
Scalene	3 cm, 4 cm, 5 cm

Equilateral: all 3 sides equal.
Isosceles: at least 2 sides equal.
Scalene: no sides equal.

Judge the measured sides, not the direction the triangle points. Under the "at least two" definition, equilateral triangles are also isosceles.

A triangle can have two names

I can... classify a triangle by angles and by side lengths.



Right isosceles

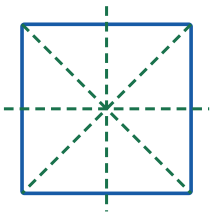
One right angle → right triangle.
Two equal sides → isosceles triangle.

The angle name and side name describe different attributes. Other angle categories: acute (3 acute angles) and obtuse (1 obtuse angle).

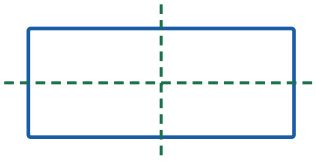
Lines of symmetry

I can… identify and draw lines of symmetry by matching halves.

Fold so both halves match



Square: 4 lines



Non-square rectangle: 2

Equal areas are not enough

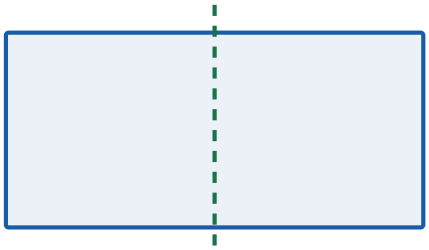
A diagonal splits a non-square rectangle into equal-area triangles, but they do not match when folded along that diagonal. It is not a symmetry line.

Some figures have none

A scalene triangle has 0 lines of symmetry. Test by matching corresponding points, not by guessing.

Test a line of symmetry by folding

I can... decide whether a line divides a figure into matching halves.



Fold here

Fold across the vertical line. The edges and corners match exactly, so it is a line of symmetry.

This rectangle has 2 symmetry lines: vertical and horizontal. A diagonal does not work for this nonsquare rectangle.

Expressions record a situation

I can... write, explain, and evaluate a two-step numerical expression.

Use parentheses for a grouped amount

Three teams each have 7 students and 4 adults.

$3 \times (7 + 4) = 3 \times 11 = 33$ people

Compare the meaning

$3 \times 7 + 4 = 21 + 4 = 25$

This describes 3 groups of 7, then only 4 more people. It is a different situation.

Explain the structure

Parentheses group what belongs in each team before multiplication.

Properties explain equivalent expressions

I can... explain how operation properties preserve the value of an expression.

Change order or grouping

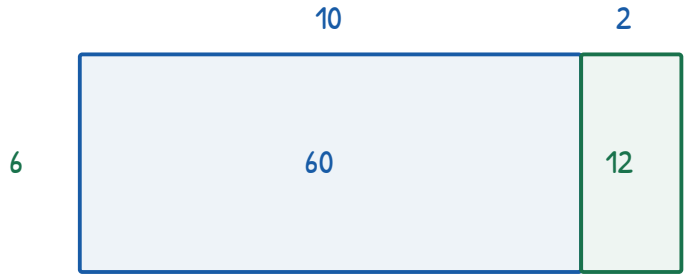
$4 \times 9 = 9 \times 4$

$(2 \times 3) \times 5 = 2 \times (3 \times 5)$

Distribute to equal parts

$6 \times (10 + 2) = 60 + 12 = 72$

Match a model



Both parts use the same factor 6. The two rectangles together make the full product.

Parentheses change which quantity comes first

I can... explain how parentheses change an expression.

Compare $4 \times (6 + 2)$ and $(4 \times 6) + 2$.

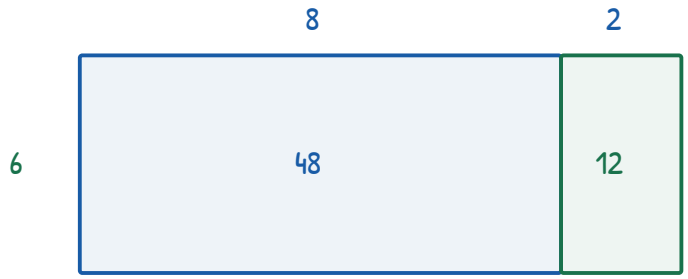
$4 \times (6 + 2) = 4 \times 8 = 32$

$(4 \times 6) + 2 = 24 + 2 = 26$

The first expression makes 4 groups of all 8. The second adds 2 only once. Each expression has two operations.

The distributive property shows equal parts

I can... explain a multiplication expression by decomposing a factor.



$6 \times (8 + 2) = 6 \times 10 = 60$

The area model has 48 squares in one part and 12 in the other.

$48 + 12 = 60$

The model explains why the two parts equal the whole product.

Compare expressions without calculating everything

I can… explain a relationship between two expressions.

$A = 8 + 7$

$B = 3 \times (8 + 7)$

B contains 3 copies of the entire sum in A. It is 3 times the value of A.

$A = 15$; $B = 45$

45 is 3 times 15. Parentheses keep the sum together as one quantity.

Two patterns make ordered pairs

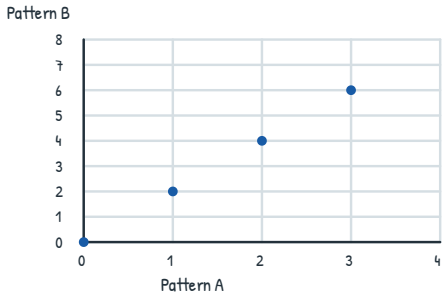
I can… generate two patterns, form ordered pairs, and explain their relationship.

Start both patterns at zero

Pattern A: add 1. Pattern B: add 2.

A	B	Ordered pair
0	0	(0, 0)
1	2	(1, 2)
2	4	(2, 4)
3	6	(3, 6)

Graph corresponding terms



Explain

Each B term is twice the corresponding A term. The points follow that relationship.

Compare corresponding terms in two patterns

I can... explain a relationship between two given rules.

Position	Start 0; add 2	Start 3; add 2
0	0	3
1	2	5
2	4	7
3	6	9

(0,3), (2,5), (4,7), (6,9)

Use a term from the first pattern for x and the corresponding term from the second for y.

Every second-pattern term is 3 greater. The rules add the same amount, so the gap stays 3.

Place value extends into decimals

I can… explain ten-times and one-tenth relationships between adjacent places.

Same digit in three places

Digit position	Value in 55.5
Tens	50
Ones	5
Tenths	0.5

Compare adjacent values

$50 = 10 \times 5 \cdot 5 = 10 \times 0.5$

$5 = 1/10 \times 50$

Name the relationship

Moving one place right makes a digit's value one tenth as much. Moving one place left makes its value ten times as much.

Multiply and divide by powers of ten

I can... explain place-value patterns when scaling by ten or one hundred.

Each digit takes a new value

$3.45 \times 10 = 34.5$

$3.45 \times 100 = 345$

Make the value smaller

$3.45 \div 10 = 0.345$

$3.45 \times 0.1 = 0.345$

Explain the digits

In 3.45, the 3 means 3 ones. After multiplying by 10, it means 3 tens. The value changed; we did not simply attach a zero.

Multiplying by 0.1 or 0.01 makes smaller units

I can... explain multiplying by one tenth and one hundredth.

Expression	Meaning
48×0.1	one tenth of 48
48×0.01	one hundredth of 48

$48 \times 0.1 = 4.8$

$48 \times 0.01 = 0.48$

Each digit's value becomes one tenth or one hundredth as much. "Add zeros" does not explain this relationship.

Divide by 10 or 100 using place value

I can... explain how division by 10 or 100 changes value.

Number	÷ 10	÷ 100
630	63	6.3
6.3	0.63	0.063

Dividing by 10 makes every value one tenth as much. Dividing by 100 makes it one hundredth as much.

$0.063 \times 100 = 6.3$

Use the inverse relationship to check.

Read and write decimals to thousandths

I can… represent a decimal in numeral, word, fraction, and expanded forms.

Read 12.407

Twelve and four hundred seven thousandths.

12.407 = 12 + 407/1,000

Expand by value

10 + 2 + 0.4 + 0.007

4 tenths; 0 hundredths; 7 thousandths.

Zero keeps a place

12.407 ≠ 12.47

In 12.47 the 7 counts hundredths, not thousandths.

Compare decimals to thousandths

I can... justify a decimal comparison using place value.

Line up units of equal value

Ones	Tenths	Hundredths	Thousandths
0	6	7	0
0	6	0	7

Find the first difference

0.670 > 0.607

Explain

Both have 6 tenths. The first number has 7 hundredths; the second has none. Thousandths cannot outweigh that difference.

0.67 = 0.670

Zeros in decimals: placeholder or equal value?

I can... explain the job of zero in a decimal.

Decimal	Read it
0.5	5 tenths
0.50	50 hundredths
0.05	5 hundredths
0.005	5 thousandths

0.5 = 0.50 = 0.500

Ending zeros can rename the same amount in smaller units.

0.5 > 0.05 > 0.005

A zero before the 5 puts it in a different place. Read the unit before comparing.

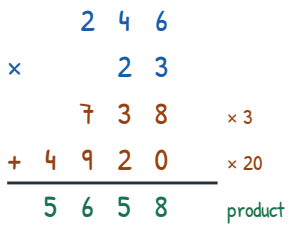
Multiply with the standard algorithm

I can... fluently multiply a three-digit number by a two-digit number and explain the steps.

246 × 23

Multiplier part	Partial product
3 ones × 246	738
2 tens × 246	4,920
Total	5,658

Record the standard algorithm



246 × 23 = 5,658

Check reasonableness

250 × 20 = 5,000

The exact answer is a little greater, as expected.

Explain the two rows in a multiplication algorithm

I can… connect algorithm rows to place-value partial products.

Multiply 326 by 24	Partial product
326 × 4 ones	1,304
326 × 2 tens	6,520

1,304 + 6,520 = 7,824

326 × 24 = 7,824

300 × 24 = 7,200

The exact answer is greater because 326 is greater than 300. The second row multiplies by 20, not 2.

Divide by a two-digit divisor

I can... use partial quotients to divide and explain a remainder.

Divide 1,725 by 24

Subtract groups of 24	Partial quotient
1,725 – 1,200 = 525	50
525 – 480 = 45	20
45 – 24 = 21	1

Combine the groups

1,725 ÷ 24 = 71 R21

Check and interpret

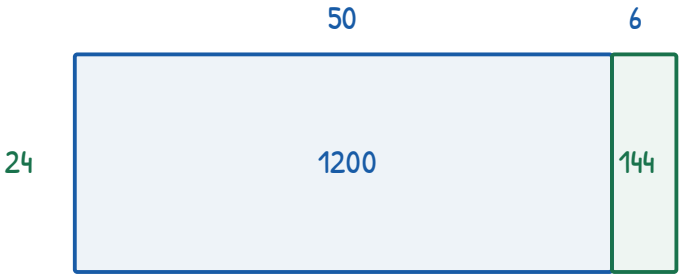
24 × 71 + 21 = 1,725

If 1,725 people need buses with 24 seats, 72 buses are needed.

An area model connects multiplication and division

I can... find a quotient by decomposing a dividend.

Find 1,344 ÷ 24. A rectangle has height 24 and area 1,344.



24 × 50 = 1,200; 24 × 6 = 144

1,344 ÷ 24 = 50 + 6 = 56

24 × 56 = 1,344

Each partial quotient is a part of the missing width.

A remainder still needs a decision

I can… interpret a remainder in a whole-number division story.

Pack 1,250 cards in boxes that each hold 24 cards.

$24 \times 50 = 1{,}200$; 50 cards remain.

$24 \times 2 = 48$; 2 cards remain.

$1{,}250 \div 24 = 52\ R2$

52 full boxes and 2 loose cards.
To hold every card in a box, use 53 boxes.

$24 \times 52 + 2 = 1{,}250$

Add decimals by like units

I can... use place value to add decimals through thousandths.

Combine matching units

2.375 + 1.48 = 2.375 + 1.480

Ones	Tenths	Hundredths	Thousandths
2	3	7	5
1	4	8	0

Regroup units

7 hundredths + 8 hundredths = 15 hundredths.
15 hundredths = 1 tenth + 5 hundredths.

2.375 + 1.480 = 3.855

Estimate

2.4 + 1.5 ≈ 3.9

Subtract decimals by renaming units

I can... rename decimal units to subtract through thousandths.

Rename 4.200

4.200 = 3 ones + 11 tenths + 9 hundredths + 10 thousandths

Subtract 1.375

(3 – 1) + (11 – 3) tenths

+ (9 – 7) hundredths + (10 – 5) thousandths

4.200 – 1.375 = 2.825

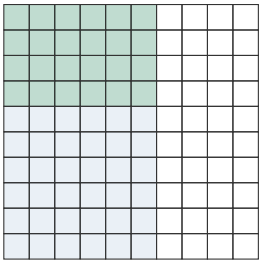
Check

2.825 + 1.375 = 4.200

Multiply decimals with unit reasoning

I can… use an area model to explain a decimal product.

0.6 by 0.4 of one whole



24 overlap parts out of 100

Count the overlap

6 tenths × 4 tenths = 24 hundredths

0.6 × 0.4 = 0.24

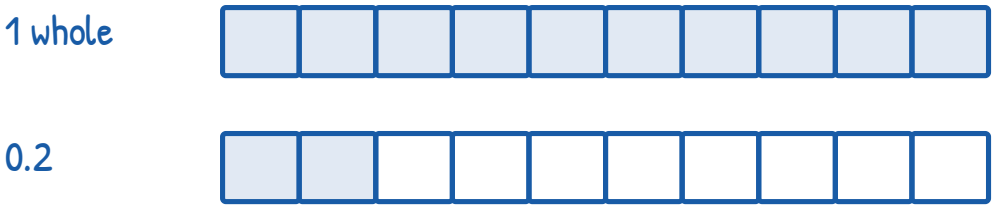
Check the size

Taking four tenths of six tenths gives less than six tenths. The product is smaller than either positive factor here.

Decimal divided by a whole number

I can... explain decimal division as equal sharing.

Share 1.2 liters among 3 bottles



There are 12 tenths altogether.

Share the tenths equally

12 tenths ÷ 3 = 4 tenths

Record and check

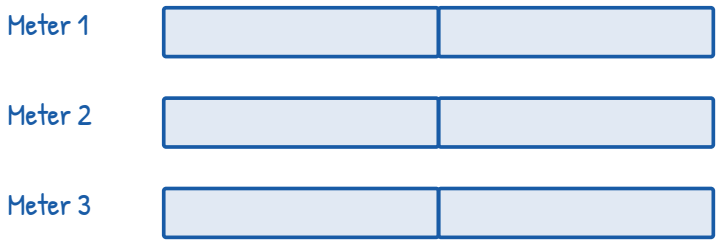
1.2 ÷ 3 = 0.4 L per bottle

3 × 0.4 = 1.2

Whole number divided by a decimal

I can... use equal groups to divide a whole number by a decimal.

How many 0.5-meter pieces in 3 meters?



Count the half-meter lengths

3 ÷ 0.5 = 6 pieces

Check the meaning

6 × 0.5 = 3 meters

The quotient counts pieces, not meters. A smaller piece size makes more pieces.

Multiply decimals with partial products

I can... use place-value parts to explain a decimal product.

Find 0.3 of 0.14 meter.

Part	0.3 of the part
0.1	0.03
0.04	0.012

$0.3 \times 0.14 = 0.03 + 0.012$

0.042 meter

Three tenths of an amount is smaller than the whole amount. 0.042 is less than 0.14.

Add decimals to thousandths in context

I can… add decimal measurements by combining like units.

Two lengths are 1.275 m and 0.438 m.

Ones	Tenths	Hundredths	Thousandths
1	2	7	5
0	4	3	8

5 + 8 = 13 thousandths → 1 hundredth + 3 thousandths.
7 + 3 + 1 = 11 hundredths → 1 tenth + 1 hundredth.

1.275 + 0.438 = 1.713 m

1.3 + 0.4 ≈ 1.7 m

The exact sum is close to the estimate.

Subtract decimal units across zeros

I can… rename decimal units without changing their value.

Find 2.000 − 0.375.

Ones	Tenths	Hundredths	Thousandths
2	0	0	0
1	9	9	10

Rename 2 as 1 one + 9 tenths + 9 hundredths + 10 thousandths.

2.000 − 0.375 = 1.625

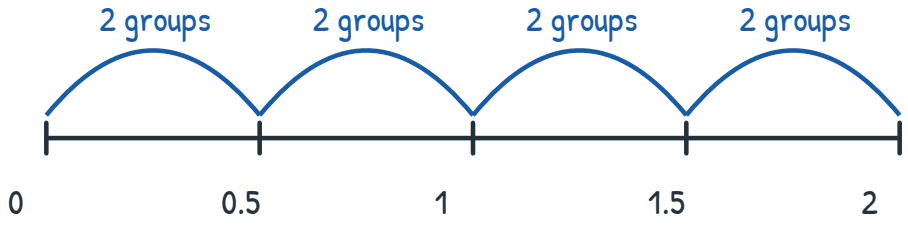
1.625 + 0.375 = 2.000

Every exchange preserves the original 2 wholes.

Repeated subtraction explains decimal division

I can… count decimal-size groups in a whole number.

How many 0.25-liter servings are in 2 liters?



Each 0.5-liter jump contains two 0.25-liter servings.

$2 \div 0.25 = 8$ servings

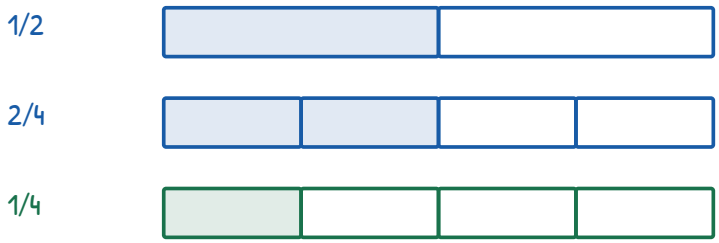
$8 \times 0.25 = 2$ liters

Division counts the groups; the group size is one fourth of a liter.

Add fractions with related denominators

I can… rename related fractions to add unlike denominators.

Make same-size parts



1/2 + 1/4 = 2/4 + 1/4 = 3/4

Explain the unit

The original units were different. Renaming the half as two fourths lets us count fourths together.

Estimate first

Half plus a quarter should be between half and one whole. Three fourths fits.

Subtract mixed numbers with related parts

I can… rename related denominators and a whole to subtract mixed numbers.

Start with 3 1/2 – 1 3/4

3 1/2 = 3 2/4 = 2 6/4

Subtract matching units

2 6/4 – 1 3/4 = 1 3/4

Explain and check

One whole becomes four fourths. Its value stays the same.

1 3/4 + 1 3/4 = 3 1/2

Two-step fraction word problems

I can… represent and solve a two-step fraction problem.

Read the connected quantities

A ribbon is 2 meters long. I use 1/2 meter, then 3/4 meter. How much remains?

Find the total used

$1/2 + 3/4 = 2/4 + 3/4 = 5/4$

Find the amount left

$2 - 5/4 = 8/4 - 5/4 = 3/4$ meter

Estimate

About 1 1/4 meters was used, so about 3/4 meter remains.

Choose related common denominators

I can… rename related fractional units before adding or subtracting.

Units in the problem	Useful shared unit
Thirds and sixths	Sixths
Fourths and eighths	Eighths
Fifths and tenths	Tenths

1/3 + 1/6 = 2/6 + 1/6

= 3/6 = 1/2

Thirds and sixths are different-size pieces. Partitioning each third into two sixths lets us combine them.

Add mixed numbers with related denominators

I can… rename and combine whole and fractional units.

Two boards measure 1 3/4 m and 2 3/8 m.

1 3/4 = 1 6/8

1 6/8 + 2 3/8 = 3 9/8

= 4 1/8 m

9 eighths include 1 whole and 1 eighth. The sum is a little more than 4 meters.

Subtract fractions using related parts

I can... use equivalent fractions to subtract unlike related units.

5/6



1/3



5/6 – 1/3 = 5/6 – 2/6

= 3/6 = 1/2

1/2 + 1/3 = 3/6 + 2/6 = 5/6

The original whole and the remaining whole have the same unit size.

Estimate a fraction answer before calculating

I can... use benchmarks to check a fraction sum.

Estimate 7/8 + 3/8. Seven eighths is close to 1. Three eighths is close to 1/2.

The sum is about 1 1/2.

Both addends are a little below their benchmarks, so the actual sum is a little below 1 1/2.

Use this estimation habit before an exact related-denominator task, such as 7/8 + 3/8 = 1 1/4.

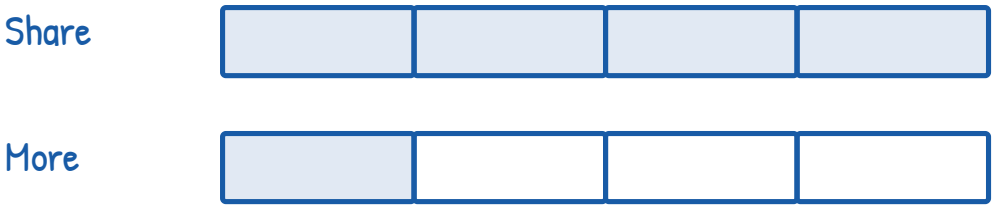
A fraction represents division

I can... model equal sharing and express the result as a fraction or mixed number.

Share 5 sandwiches among 4 people

Cut every sandwich into four equal pieces. Each person gets one fourth from each sandwich.

Count one person's share



5 ÷ 4 = 5/4 = 1 1/4 sandwiches

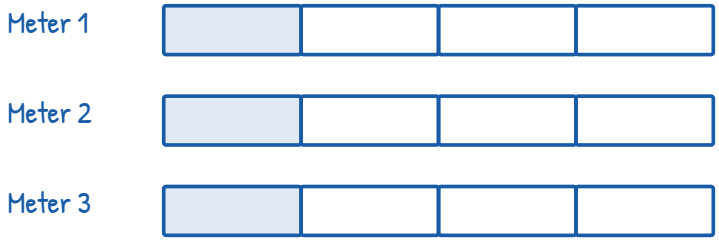
Connect the numerator and denominator

5 is the amount shared. 4 is the number of equal shares. The fraction names the amount in each share.

Equal sharing can give less than one whole

I can... model whole-number division with a fractional quotient.

Share 3 meters of ribbon equally among 4 students.



Each student gets 1/4 meter from each of the 3 meters.

3 ÷ 4 = 3/4 meter each

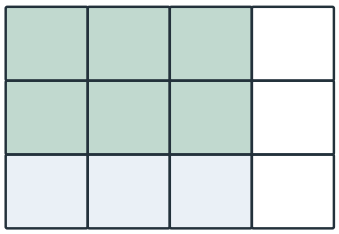
4 × 3/4 = 3 meters

Blue shows one student's total share.

Multiply fractions using overlap

I can... use an area model to explain multiplying two fractions.

Find 2/3 of 3/4



6 overlap parts out of 12

Count overlap parts

2/3 × 3/4 = 6/12 = 1/2

Explain the units

The whole is divided into 12 equal parts. Three fourths selects 3 columns. Two thirds selects 2 rows of those columns. Six parts overlap.

Multiply with mixed numbers

I can... decompose a mixed number to explain a fraction product.

Find half of 1 1/2

1/2 × (1 + 1/2)

Take half of each part

(1/2 × 1) + (1/2 × 1/2)

Half of 1



Half of 1/2



Combine the amounts

1/2 + 1/4 = 3/4

Multiplication can shrink or grow

I can... predict how a positive multiplier changes a number's size.

Take less than one copy

$8 \times 1/2 = 4 < 8$

Take exactly one copy

$8 \times 1 = 8$

Take more than one copy

$8 \times 1\ 1/2 = 12 > 8$

Explain

For a positive starting number, a multiplier less than 1 takes part of it. A multiplier greater than 1 takes more than one copy.

A fraction of a whole-number amount

I can… model a fraction multiplied by a whole number.

Find 3/4 of 8 cups.



Partition 8 into 4 equal shares. Each share is 2 cups. Take 3 shares.

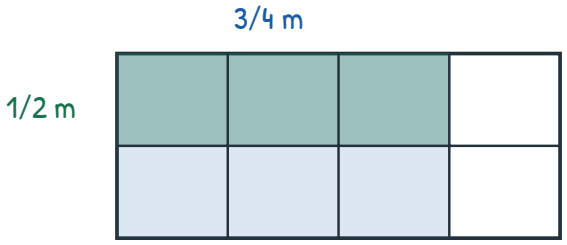
$3/4 \times 8 = 3 \times 2 = 6$ cups

6 is less than 8 because three fourths is less than one whole.

Fractional side lengths create fractional area

I can... use area to explain a fraction product.

A rectangle is 3/4 meter wide and 1/2 meter tall.



3 of 8 equal parts = 3/8 m²

split a 1 m by 1 m whole into fourths across and halves down. The green rectangle covers 3 of the 8 equal parts.

$3/4 \times 1/2 = 3/8$ m²

The product counts square meters, not meters. A half of the strip is smaller than the whole strip.

Divide a whole number by a unit fraction

I can... count unit-fraction groups in a whole-number amount.

Cut 2 meters into quarter-meter pieces

Meter 1 

Meter 2 

Count the pieces

$2 \div 1/4 = 8$ pieces

Check with multiplication

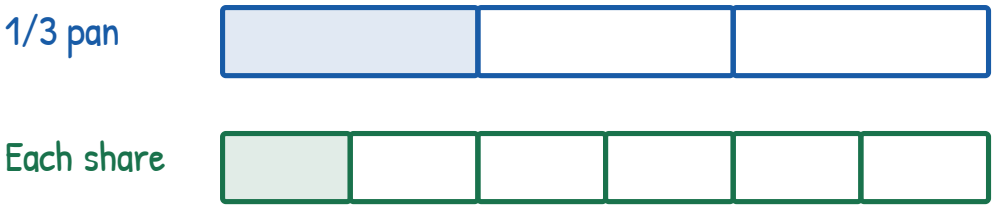
$8 \times 1/4 = 2$

There are four quarter-meter pieces in each meter. Two meters contains eight pieces.

Divide a unit fraction into equal shares

I can... divide a unit fraction by a whole number using a model.

Share 1/3 of a pan among 2 people



Split the third in half

1/3 ÷ 2 = 1/6 of the whole pan

Check

2 × 1/6 = 1/3

Each person gets half of the available third, not half of the entire pan.

Check unit-fraction division with multiplication

I can... explain and check either type of unit-fraction division.

Question	Equation
How many fourths are in 3?	3 ÷ 1/4 = 12
Share 1/4 equally among 3.	1/4 ÷ 3 = 1/12

12 × 1/4 = 3

3 × 1/12 = 1/4

The first answer counts pieces. The second names the size of each share. The same numbers describe different stories.

Convert within a measurement system

I can... use a provided conversion chart to solve a conversion problem.

Use the given relationship

Given	Conversion
1 foot = 12 inches	2.5 feet = ? inches
1 meter = 100 centimeters	350 centimeters = ? meters

Multiply for smaller units

2.5 × 12 = 30 inches

Divide for larger units

350 ÷ 100 = 3.5 meters

Time conversion uses the given relationship

I can... use a conversion chart to convert a time measure.

Given	Find
1 minute = 60 seconds	2 1/2 minutes = ? seconds

2 × 60 = 120; 1/2 × 60 = 30

2 1/2 minutes = 150 seconds

Seconds are smaller units, so the numerical count is greater. The length of time stays the same.

Choose a graph for the data

I can… distinguish categorical, numerical, and changing-over-time data.

Ask what the answers represent

Question	Data
Favorite fruit?	Categorical
Number of pets?	Numerical
Plant height each week?	Changes over time

Match a representation

A bar graph compares categories. A line plot shows frequencies of numerical values. A line graph shows how a measured quantity changes over time.

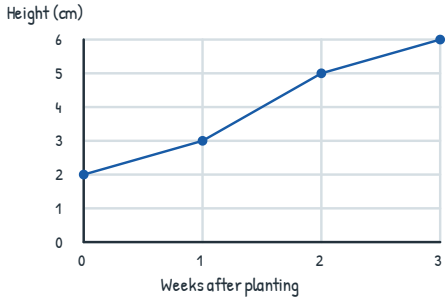
Plan a time-based question

"How does our plant's height change each week?" Record the date and height each time.

Read a line graph over time

I can… create a line graph and describe a change over time.

Plant growth



Read the coordinates in context

At week 2, the plant is 5 cm tall.

Find and compare changes

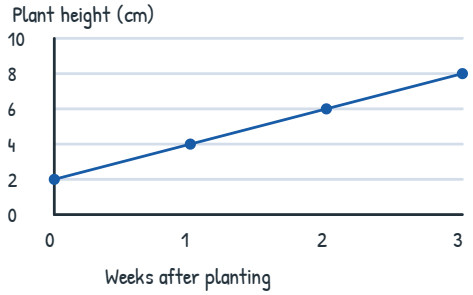
Week 1 → 2: 5 − 3 = 2 cm

Week 2 → 3: 6 − 5 = 1 cm

The plant grew more during the first of these intervals.

Create a line graph from a time-based table

I can… choose a scale and graph data that change over time.

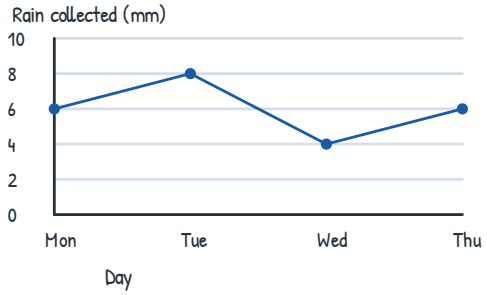


Data: week 0 → 2 cm; week 1 → 4 cm; week 2 → 6 cm; week 3 → 8 cm.
Plot each pair, then connect observations in time order.

Time belongs on the horizontal axis. The height scale counts by 2 cm. Label the title, axes, and units.

Describe increases and decreases on a line graph

I can… compare changes between consecutive observations.



Mon → Tue: increase of 2 mm

Tue → Wed: decrease of 4 mm

Wed → Thu: increase of 2 mm

The greatest change is the decrease from Tuesday to Wednesday. A higher point tells the amount; a segment tells the change.

Volume counts unit cubes

I can… count layers of unit cubes to measure volume.

One layer contains 6 unit cubes



3 columns × 2 rows

Stack 4 identical layers

6 + 6 + 6 + 6 = 24 cubic units

Name the unit

Each cube is 1 unit long, 1 unit wide, and 1 unit high. Volume counts space filled, not just the visible faces.

Choose a cubic unit for volume

I can... explain why volume is measured in cubic units.

Measure	Unit measures
Length: cm	One direction
Area: cm^2	A 1 cm by 1 cm square
Volume: cm^3	A 1 cm by 1 cm by 1 cm cube

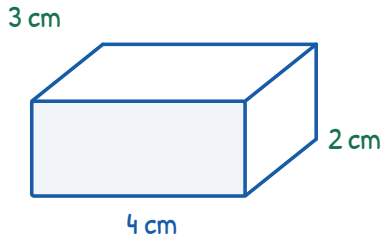
Pack cubes into a solid without gaps or overlaps. Count every cube, including cubes hidden behind others.

A box with 24 one-centimeter cubes has volume 24 cm^3. Counting only visible faces measures the wrong thing.

Build the volume formula from layers

I can… explain why multiplying three edge lengths gives a rectangular prism's volume.

A 4 cm by 3 cm by 2 cm prism



Find cubes in each layer

4 × 3 = 12 cubes per layer

Multiply by the number of layers

12 × 2 = 24 cm^3

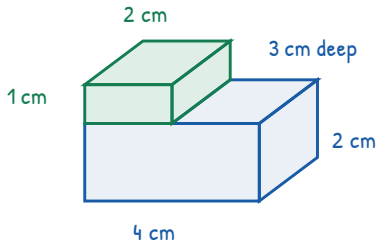
V = length × width × height

Add volumes of nonoverlapping prisms

I can… decompose a solid into two prisms and add their volumes.

Split a step-shaped solid

A lower prism is 4 cm × 3 cm × 2 cm. A top prism is 2 cm × 3 cm × 1 cm. The top sits on the lower prism.



Blue: lower prism · Green: upper prism

Find each part

Lower: $4 \times 3 \times 2 = 24$ cm³

Top: $2 \times 3 \times 1 = 6$ cm³

Add without double counting

$24 + 6 = 30$ cm³

The prisms touch but their interiors do not overlap.

Base area times height counts all the layers

I can… connect base area and height to prism volume.

A prism has base area 15 cm² and height 4 cm.

One layer	Number of layers
15 unit cubes	4

$V = B \times h$

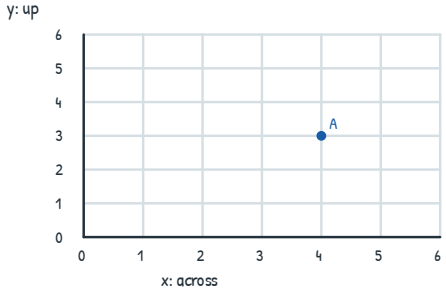
$V = 15 \times 4 = 60 \text{ cm}^3$

B names base area. Each 1-cm-high layer has 15 cubes. Four identical layers fill the solid.

Plot and interpret ordered pairs

I can… plot a point in the first quadrant and interpret its coordinates.

Start at the origin



Read (4, 3) in order

From (0, 0), move 4 units right, then 3 units up. The first coordinate gives x; the second gives y.

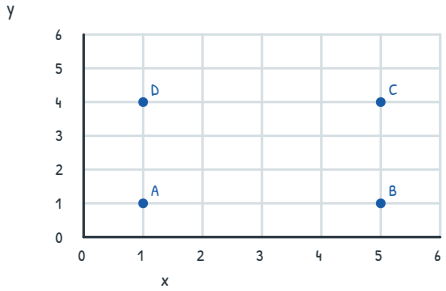
Interpret the point

If x counts tickets and y counts dollars, (4, 3) describes 4 tickets costing 3 dollars. Axis labels give the numbers meaning.

Find missing coordinates in a shape

I can... use coordinates to solve a geometric problem.

Complete a rectangle



Match vertical and horizontal positions

A = (1, 1), B = (5, 1), C = (5, 4).
D shares A's x-coordinate and C's y-coordinate.

D = (1, 4)

Check the side lengths

Width: 5 – 1 = 4 units

Height: 4 – 1 = 3 units

Points can lie on an axis or at the origin

I can… interpret a zero coordinate in the first quadrant.

Point	How to locate it
(0, 0)	Stay at the origin
(0, 4)	0 right, then 4 up
(5, 0)	5 right, then 0 up

If x = 0, the point is on the y-axis.
If y = 0, the point is on the x-axis.

The order matters. (0, 4) and (4, 0) are different locations. Include the axes when describing first-quadrant graphs.

Quadrilateral categories form a hierarchy

I can… explain how a quadrilateral belongs to more than one category.

Start with inherited attributes

Quadrilateral → four sides.
Parallelogram → two pairs of parallel sides.

Two branches can overlap

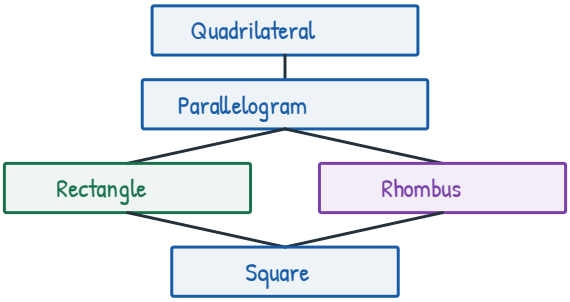
Category	Extra attribute
Rectangle	4 right angles
Rhombus	4 equal sides
Square	Both attributes

Explain the hierarchy

Every square is a rectangle and a rhombus. Both are parallelograms, so every square is also a parallelogram and a quadrilateral. The reverse is not always true.

A visual hierarchy of quadrilaterals

I can... explain inherited attributes in a quadrilateral hierarchy.



A square has 4 right angles and 4 equal sides. It belongs to both the rectangle and rhombus categories.

All squares are rectangles. Some rectangles are not squares. Properties from broader categories still apply.

Sort quadrilaterals using attributes

I can... justify a shape's categories using its properties.

Shape has...	It belongs to...
4 equal sides	Rhombus
4 right angles	Rectangle
Both of these	Square

A rhombus with an obtuse angle is not a rectangle. A square is a rhombus and a rectangle.

Check attributes even when a shape is tilted. Orientation and apparent size do not decide its category.